

Sudesh Subedi

Computer Engineer,
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Education

TRIBHUVAN UNIVERSITY

IOE, Purwanchal Campus, Dharan, Nepal

Bachelor in Computer Engineering, 2022-2026

Relevant Courses:

Data Structures and Algorithms, Operating Systems, Database Management System, Computer Organization and Architecture, Computer Networks, Software Engineering, Project Management

Major Project: Implementing Read/Write Overlay Filesystem Support in a Microkernel Based Operating System

Supervisor: Er. Manoj Kumar Guragai

Minor Project: A Decentralized and Federated Social Media for Scientific Communications.

Supervisor: Er. Pukar Karki

CAPITAL COLLEGE AND RESEARCH CENTER

Koteshwore, Kathmandu

National Examination Board +2, Science, 2019-2021

Projects

2025-2026 **A Read/Write Overlay Filesystem Support in a Microkernel Based Operating System** *Purwanchal Campus, Dharan, Nepal*

Currently working on developing Overlay(Union) File System for Redox OS. It combines multiple different mount points of an existing filesystem into one, resulting in a single directory structure that contains underlying files and sub-directories from all sources.

2024-2025 **A Federated Social Media for Scientific Communicators** *Purwanchal Campus, Dharan, Nepal*

Developed 3o14 (pi), a single user ActivityPub based fediverse client and server that supports rendering mathematical typesetting as a way to scientifically communicate between users, facilitating to write posts and replies using Markdown and LaTeX with a complete microblogging experience.

2024 **A UCI Compatible Chess Engine** *Zig*

Developed a UCI-compatible chess engine in Zig with support for core UCI commands, move generation, board evaluation, and game state management. Focused on low-level systems programming, engine architecture, and search logic while exploring performance-oriented design in Zig.

Mini Projects

RustyJP A CLI tool for learning Japanese alphabets (Hiragana and Katakana) as well as Kanji using spaced repetition method and custom kanji addition.

Zhip-8 CHIP-8 emulator written in zig featuring opcode decoding, memory management, display rendering, timers, and input handling.

PngMe Rust-based CLI application for PNG steganography, supporting message encoding/decoding, chunk listing, and safe removal of embedded data within PNG file structure.

Research Interests

Operating Systems, Distributed Systems, Advance Algorithms and Data Structures, Compiler Development

Extras

Spoken Languages

English (*Academic*), Nepali (*Native*), Hindi(*Conversational*)